# GREY HUNTER

## CYBERSECURITY TOOLS FIELD HANDBOOK & OPERATIONAL MANUAL

Comprehensive Technical Reference • 22+ Security Auditing & Reconnaissance Tools
Downloadable Field Manual • Offline Ready • Ethical Hacking Edition

## 1. Nmap // Network Mapper

**Category:** NETWORK RECON

**Overview:** The industry-standard port scanner and network exploration tool. Knocks on TCP/UDP ports to detect listening daemons, active services, and operating systems.

**Under The Hood Mechanics:** Sends crafted TCP SYN, UDP, and ICMP echo probes. Analyzes minute TCP window size and flag variations to fingerprint host operating systems.

| Flag | Description |
|---|---|
| **-sS** | TCP SYN Stealth Scan. Sends SYN without completing 3-way handshake. |
| **-sV** | Version Detection. Probes open ports to identify software versions. |
| **-p** | Port Specification (-p 1-65535 or -p 80,443). |
| **-A** | Aggressive scan (OS detection, versioning, default scripts). |

### Ready-To-Run Command:

```
nmap -sS -sV -T4 -p 22,80,443,8080 192.168.1.100
```

**Defensive Hardening Tip:** Configure firewalls to drop unexpected inbound SYN packets, close unused ports, and deploy Snort/Suricata IDS.

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## 2. Netdiscover // Live ARP Scanner

**Category:** NETWORK RECON

**Overview:** Active and passive Address Resolution Protocol (ARP) reconnaissance tool designed for wireless networks and local subnets.

**Under The Hood Mechanics:** Broadcasts ARP 'Who has IP X?' requests. Because all network devices must answer ARP queries to communicate on local LANs, host firewalls cannot block them.

| Flag | Description |
|---|---|
| **-r** | Subnet range to scan (e.g. -r 192.168.1.0/24). |
| **-i** | Network interface to use (-i wlan0, eth0). |
| **-p** | Passive mode. Sniffs silently without transmitting any packets. |

### Ready-To-Run Command:

```
netdiscover -i wlan0 -r 192.168.1.0/24
```

**Defensive Hardening Tip:** Deploy Dynamic ARP Inspection (DAI) and 802.1X Network Access Control (NAC) on managed switches.

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## 3. RustScan // Modern Fast Port Scanner

**Category:** NETWORK RECON

**Overview:** Modern asynchronous port scanner written in Rust that scans all 65,535 ports in under 3 seconds, automatically piping findings into Nmap.

**Under The Hood Mechanics:** Employs an adaptive socket scanning algorithm that dynamically scales concurrency based on network latency and error rates.

| Flag | Description |
| --- | --- |
| **-a** | Target IP address or CIDR range. |
| **-r** | Port range to scan (-r 1-65535). |
| **-b** | Batch size for concurrent sockets (-b 4500). |

### Ready-To-Run Command:

```
rustscan -a 192.168.1.100 -- -sV
```

**Defensive Hardening Tip:** Configure firewall connection rate-limiting to automatically ban IPs initiating hundreds of connections per second.

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## 4. Nuclei // Template-Based Vulnerability Scanner

**Category:** WEB VULNERABILITY

**Overview:** Fast, community-driven vulnerability scanner based on simple YAML templates for web, cloud, and network assessments.

**Under The Hood Mechanics:** Executes declarative YAML templates defining exact HTTP, DNS, TCP, or SSL requests and matches strings/regexes to verify CVEs.

| Flag | Description |
|---|---|
| **-u** | Target URL to scan (-u https://target.com). |
| **-s** | Severity filter (-s critical,high,medium). |
| **-tags** | Filter by category tags (-tags cve,auth,exposure). |

### Ready-To-Run Command:

```
nuclei -u https://target.com -s critical,high
```

**Defensive Hardening Tip:** Integrate Nuclei into CI/CD deployment pipelines to detect exposed configs and known CVEs before code reaches production.

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## 5. OWASP Amass // Attack Surface Mapping

**Category:** OSINT & RECON

**Overview:** In-depth external asset discovery and attack surface mapping tool using open-source intelligence and active DNS resolution.

**Under The Hood Mechanics:** Combines passive scraping of Certificate Transparency logs, Shodan, and Censys with active recursive DNS lookups and ASN routing analysis.

| Flag | Description |
|---|---|
| **enum** | Subdomain enumeration mode. |
| **-active** | Active probing of discovered hosts. |
| **-brute** | Brute-force subdomains using wordlists. |

### Ready-To-Run Command:

```
amass enum -passive -d target.com
```

**Defensive Hardening Tip:** Continuously monitor Certificate Transparency logs and decommission legacy unused subdomains to prevent Subdomain Takeovers.

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## 6. Subfinder // Fast Passive Subdomain Finder

**Category:** OSINT & RECON

**Overview:** Ultra-fast passive subdomain discovery tool that queries over 30 public sources without sending a single packet to the target.

**Under The Hood Mechanics:** Queries certificate search engines, web archives, and DNS repositories via public APIs, leaving zero trace on the target server.

| Flag | Description |
|---|---|
| **-d** | Domain to enumerate (-d target.com). |

| | |
|---|---|
| **-silent** | Only print clean discovered subdomains. |
| **-all** | Query all passive sources. |

### Ready-To-Run Command:

```
subfinder -d target.com -silent
```

**Defensive Hardening Tip:** Restrict DNS zone transfers (AXFR) and monitor public domain registries for unauthorized lookalike domains.

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## 7. WPScan // WordPress Security Auditor

**Category:** WEB CMS AUDIT

**Overview:** Black-box security scanner that catalogs installed WordPress core files, plugins, and themes, correlating them against the WPScan Vulnerability Database.

**Under The Hood Mechanics:** Scans HTML headers, CSS links, and readme paths to detect exact component versions and checks for unpatched vulnerabilities.

| Flag | Description |
|---|---|
| **--url** | Target WordPress site URL. |
| **-e vp** | Enumerate vulnerable plugins. |
| **-e u** | Enumerate administrative usernames. |

### Ready-To-Run Command:

```
wpscan --url https://target.com -e vp,u
```

**Defensive Hardening Tip:** Disable the WordPress REST API users endpoint, enable automated security updates, and enforce 2-Factor Authentication.

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## 8. TruffleHog // Secret & Credential Leak Detector

**Category:** SECURITY AUDITING

**Overview:** Scans Git commit histories and filesystems to find exposed API tokens, AWS secret keys, database passwords, and private keys.

**Under The Hood Mechanics:** Combines Shannon entropy calculation with 700+ regex token detectors and performs live verification against provider APIs.

| Flag | Description |
|---|---|
| **git** | Scan a Git repository commit history. |
| **filesystem** | Scan local source directory. |
| **--only-verified** | Output only confirmed active secrets. |

### Ready-To-Run Command:

```
trufflehog filesystem ./src --only-verified
```

**Defensive Hardening Tip:** Enforce pre-commit hooks to block sensitive secrets and credentials from ever entering version control repositories.

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